

# NAITIK JAIN

**AI/ML Engineer | Generative AI | Agentic AI | OCR & Document Intelligence**

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## PROFESSIONAL EXPERIENCE

**AI/ML Engineer | WONDRx | July 2025 – Present | Mumbai**

- Fine-tuned Microsoft TrOCR-Base Handwritten on WONDRx's proprietary dataset of handwritten patient names extracted from prescription images, achieving a 10% Character Error Rate (CER). Built and deployed a FastAPI-based inference service, containerized with Docker, currently serving doctors in production.
- Built a multi-role healthcare AI assistant using LangGraph, Gemini, and MCP, exposing 62 API-backed tools with persistent conversational memory, role-based access control, secure identity injection, in-process MCP communication, and end-to-end tool-call observability.
- Led the development of a prescription digitization system for hard-to-read handwritten prescriptions, achieving 80% accuracy while reducing manual QC effort and improving scalability. Used VLMs for OCR, LLMs for contextualized structured extraction, FastAPI, and PostgreSQL.

**Automation & AI Intern | WONDRx | Dec 2024 – Jun 2025 | Mumbai**

- Built an OCR automation pipeline for doctor onboarding, eliminating manual data entry and streamlining the onboarding process using VLM-based OCR and Flask.
- Developed an analytics dashboard for a top-8 pharmaceutical company in India, reducing reporting latency from one month to 24 hours and enabling faster decision-making. Collaborated with management and enterprise stakeholders to align the solution with business requirements using Python, Streamlit, role-based access control, and API integrations.

## RESEARCH & PUBLICATIONS

**[PodBias: Unveiling Bias in YouTube Podcasts Using Deep Learning](#)**

- Built a media bias detection system using Sentence-BERT and BART to identify and analyze bias in podcast content.
- Created a custom dataset of 4,000+ labeled sentences for training and evaluating the bias detection system.
- Deployed a Streamlit application providing real-time bias scoring and accessible content analysis.

## TECHNICAL SKILLS

**AI & Generative AI:** LLMs, VLMs, OCR, NLP, Transformers, Computer Vision, RAG, Agentic AI, LangGraph, MCP

**Backend & Databases:** FastAPI, Flask, PostgreSQL, REST APIs, Typesense, ChromaDB

**Programming:** Python, SQL

**Data & Visualization:** Streamlit, Tableau, Power BI

**Deployment & Infrastructure:** Docker, Linux

**Finance:** Financial Analysis, Quantitative Finance, Algorithmic Trading

## KEY PROJECTS

**Multimodal RAG System for IPO/DRHP Analysis**

*Python | Docling | PyMuPDF | ChromaDB | BM25 | Sentence Transformers | OpenRouter | Ollama | Streamlit*

Built an end-to-end multimodal RAG application for querying 600–700-page IPO filings and financial documents with evidence-grounded answers and page-level citations.

- **Ingestion & Indexing:** Built a resumable document-processing pipeline using Docling, PyMuPDF quality checks, selective OCR, vision-LLM chart captioning, section-aware chunking, and BGE embeddings, with vectors stored in ChromaDB alongside a BM25 index.
- **Retrieval:** Implemented metadata-filtered hybrid retrieval combining dense vector search and BM25, fused results using Reciprocal Rank Fusion (RRF), and added optional cross-encoder reranking to retrieve relevant text, tables, and visual information.
- **Generation & Interface:** Developed evidence-grounded generation using OpenRouter with local Ollama fallback, page-level citation validation to reduce hallucinations, and a Streamlit interface supporting conversational Q&A, source previews, retrieval scores, and cited PDF-page rendering.

## **AutoTrend Breakout Strategy — Algorithmic Trading**

*Python | Machine Learning | Random Forest | Gradient Descent*

- Developed a Python-based trading system using Gradient Descent to mathematically identify optimal support and resistance trendlines.
- Integrated a Random Forest classifier to validate breakout signals using volume indicators and price deviations, filtering false-positive trade signals.

## **Stock News Notification System**

*Python | Zerodha Kite Connect API | Telegram Bot API | Google News RSS | Gemini API*

- Developed an automated stock-news alert system to monitor Zerodha portfolio holdings and deliver relevant market-news notifications through Telegram.
- Integrated Zerodha Kite Connect for holdings retrieval, Google News RSS for news aggregation, and Gemini for sentiment analysis and concise article summarization.
- Implemented alert prioritization, duplicate-news filtering, watchlists, mute functionality, portfolio P&L summaries, and price-movement detection to improve signal quality.
- Built an interactive Telegram bot supporting portfolio queries, stock-specific news commands, inline stock search, and personalized alert management.

## **EDUCATION**

**B.Tech in Data Science | MPSTME, NMIMS | CGPA: 3.6/4**

**Higher Secondary | Pace Junior Science College | 90%**

**Secondary School | Ryan International School | 93.5%**

## **CERTIFICATIONS**

- Unsupervised Learning, Recommenders, Reinforcement Learning — DeepLearning.AI & Stanford University
- Amazon ML Summer School 2023
- Machine Learning with Tree-Based Models in Python — DataCamp
- Quantitative Finance with Python — Udemy
- Data Analytics A-Z with Python — Udemy
- Python Introduction to Data Science and Machine Learning — Udemy

## **VOLUNTEERING**

**Volunteer | Indian Development Foundation, Mumbai | June 2022 – July 2022**

- Supported center operations assisting patients suffering from leprosy.
- Authored a report detailing United Nations initiatives toward eradicating world hunger.
- Reference: Project Document